Combat UAV

UAV Technical Specifications

Category: The competition is held only in the **Fixed-Wing** category. Vehicles capable of **vertical takeoff and landing (VTOL)** but performing flight through wings also fall into this class.

Weight: Aircraft must weigh a maximum of **12 kg**.

Propulsion System: Only **battery-powered electric motors** may be used.

Software Originality: Ready-made guidance software cannot be used; mission software must be **originally developed by the teams**.

Safety:
Vehicles must have an **externally accessible current cut-off switch (fuse)** and **fail-safe systems** that activate in the event of communication loss.

Competition Missions

The competition consists of two main missions: **"Combat UAV"** and **"Kamikaze UAV."**
To win the grand prize, both missions must be completed successfully.

Mission Type Details

Combat UAV Mission

The goal is to **virtually lock onto rival UAVs through the camera** and to **avoid being locked onto**.
A hit is achieved by keeping the center of the rival vehicle in the camera image for at least **4 seconds**.

Kamikaze UAV Mission

The vehicle is expected to **autonomously dive toward the designated ground target** (a **2x2 m QR code**) and read the code.

A physical collision is not desired; the UAV must perform a **safe fly-by / go-around** at a safe altitude.

Air Defense and Jamming:
Virtual **no-fly zones** may be activated in the competition area; entering these zones results in penalty points.

Scoring and Evaluation

The total score is calculated by combining reports, videos, and flight missions:

- **Critical Design Report (CDR):** 15%
- **System Identification Video (SIV):** 15%
- **Mission Scoring:** 70%
  (including autonomous takeoff, landing, successful locks, and kamikaze points)

Penalties:
Incorrect lock declaration, exceeding the manual mode transition limit, and boundary violations are reflected as negative points.

Software Team (Autonomy and Decision Mechanism)

The software team develops the **brain** of the vehicle and its autonomous mission capabilities:

- **Autonomous Flight Control:** Coding the UAV's autonomous takeoff, mission flight, and landing functions.
- **Image Processing:** Detecting rival UAVs through camera images and developing **tracking algorithms** to maintain lock for at least **4 seconds**.

- **Communication Protocols:** Sharing telemetry data (GPS, altitude, etc.) and lock packets with the competition server at a rate of **1–5 Hz**.
- **Kamikaze and QR Reading:** Writing algorithms to autonomously detect and read the **2x2 m QR code** on the ground during the dive.
- **Avoidance Algorithms:** Autonomous route planning to prevent entering virtual air defense and signal jamming zones (**no-fly areas**).
- **Ground Control Station (GCS):** Designing a user interface that shows borders, rivals, and restricted zones instantly on the map.

Electrical-Electronics Team (Power and Communication Infrastructure)

This team ensures the vehicle's energy management and continuity of hardware communication:

- **Power System:** Battery selection, integration of electric propulsion motors, and planning of power distribution boards.
- **Safety Equipment:** Installation of an easily accessible external current cut-off switch and a fuse providing high-current protection.
- **RF and Signal Management:** Assignment and testing of telemetry, video transmission, and control transceiver frequencies (for example, in the **5.1–5.9 GHz** range).
- **Wiring and Safety:** Selecting all electronic cables with sufficient thickness and securing connectors against flight vibration.

- **Fail-Safe Systems:** Establishing hardware/software safety measures that shut down the motor or switch to **Return-to-Launch (RTL)** mode in case of signal loss.

Mechanical Team (Design, Aerodynamics, and Manufacturing)

The mechanical team is responsible for the physical structure and durability of the vehicle:

- **Airframe Design:** Original design and production of a **fixed-wing** body, a **VTOL-capable** body, or a body with conventional landing gear.
- **Aerodynamic Calculations:** Sizing the wings, tail, and control surfaces for aerodynamic efficiency and stability.
- **Weight Management:** Ensuring that the vehicle remains within the **maximum 12 kg takeoff weight limit** with all equipment installed.
- **Structural Integrity:** Mounting components (camera, battery, motor, etc.) securely to the body without vibration, and using nuts, adhesive, or safety wire in the mounting.
- **Camera Placement:** Positioning the lock-on camera on the **nose of the aircraft**, fixed and parallel to the horizontal axis.